

4 submit

Oscar Vargas Pabon

osvarp@javerianacali.edu.co

8975574

Actividad juego de la vida de Conway

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Este código y el video esta en https://github.com/osvarp/paralelComputing_week3

En la carpeta /4submit

EN COLAB

Intel(R) Xeon(R) CPU @ 2.20GHz

2 cpus logicas

Secuencial:

3.96 s

3.86 s

3.93 s

2.85 s

2.82 s

Promedio :

3.546

Multithreading:

```
a=[2.9336349964141846,2.956101179122925,4.029467821121216,3.697298288345337,2.9422197341918945]
```

```
a.sort()
```

```
a.pop();a.pop(0);  
print(sum(a)/len(a))
```

Promedio:

3.1985397338867188

Multiprocessing:

```
a=[2.9357314109802246,2.3459312915802,2.3314685821533203,2.3291428089141846,2.3  
34846258163452]  
  
a.sort()  
  
a.pop();a.pop(0);  
print(sum(a)/len(a))
```

Promedio:

2.3374153772989907

Hallando los spedups

```
speedup=lambda t_seq,t_ot:t_seq/t_ot  
  
t_seq=3.546; t_thrd=3.1985397338867188; t_proc=2.3374153772989907;  
  
print("Speedup thread <{0}>\nSpeedup procce  
<{0}>"%(speedup(t_seq,t_thrd),speedup(t_seq,t_proc)))
```

Respuesta:

Speedup thread <1.108631>

Speedup procce <1.517060>

MI PC

Numero de CPUs lógicas: 8

Porcentaje de uso de la CPU 16.7%

Total memory 6091.31 MB, porcentaje de memoria disponible 25.34%

Secuencial

```
a=[1.445688247680664,1.4506659507751465,1.4660518169403076,1.4744021892547607,1.7438380718231201];
```

```
a.sort();
```

```
x=a.pop();xx=a.pop(0);
```

```
print(sum(a)/len(a));
```

Promedio: 1.463706652323405

MULTITHREADING

```
a=[2.542898178100586,1.8883261680603027,1.8575701713562012,1.8412299156188965,1.923079490661621];
```

```
a.sort();
```

```
x=a.pop();xx=a.pop(0);
```

```
print(sum(a)/len(a));
```

Promedio: 1.8896586100260417

multiprocessing

```
a=[2.3536012172698975,2.2455637454986572,2.272753953933716,2.3945300579071045,2.5302860736846924];
```

```
a.sort();
```

```
x=a.pop();xx=a.pop(0);
```

```
print(sum(a)/len(a));
```

promedio: 2.3402950763702393

Hallando los spedups

```
speedup=lambda t_seq,t_ot:t_seq/t_ot;  
t_seq=1.463706652323405; t_thrd=1.8896586100260417; t_proc=2.3402950763702393;  
print("Speedup thread <%.f>\nSpeedup procce  
<%.f>"%(speedup(t_seq,t_thrd),speedup(t_seq,t_proc)));
```

Respuesta:

Speedup thread <0.774588>

Speedup procce <0.625437>